



癲癇 ICD Crosswalk Hackathon 專案摘要

固定範圍、可追溯來源、獨立藥物證據、驗證與人工審查 handoff



ICD-9-CM codes

5



Medications

3



GEM candidate rows

11



Validation

38 / 38 passed



專案目標

- 保留 5 個固定癲癇 ICD-9-CM code 的所有 GEM 候選
- 用 FY2026 2026/04/01 ICD-10-CM release 做 exact lookup
- 整理 3 種抗癲癇藥物的 disease-scope evidence
- 產出可交付的 crosswalk audit package



核心輸出

- icd_crosswalk.csv
- drug_evidence.csv
- audit_summary.csv
- source_inventory.csv
- validation_report.md
- review_note.md



Take-home message : 這不是自動選碼工具，而是一份可追溯、可驗證、需人工審查的候選 crosswalk 與 evidence audit package。



插入頁 | project_spec.md

以 md 文字呈現專案定義、範圍與完成條件



1. 問題定義

針對固定五個 ICD-9-CM 癲癇細項，保存 CMS 2018 GEMs 的全部 ICD-10-CM 候選、原始旗標、版本與來源，並以 FY 2026 ICD-10-CM April 1 release 做 exact code lookup。另對最多三種指定藥物整理可回查的疾病範圍證據。

2. 使用者情境

醫療資料工程與 evidence audit 展示；不是病人診斷、自動編碼或處方工具。

3. 固定範圍

345.00、345.10、345.40、345.90、345.91；藥物為 Phenytoin、Carbamazepine、Valproic acid。

4. In scope

候選擷取、raw flags、cardinality、target existence、藥物名稱正規化、窄範圍 evidence、validator、synthetic eval、human-review handoff。

5. Out of scope

病人層級診斷、最終 coding、申報決策、用藥推薦、劑量、安全性評估、基因證據、反向 mapping。

6. 完成條件

五碼與三藥均處理；所有輸出 schema 完整；validator 與 normal/not-found/conflicting eval 通過；來源限制明示。



Safety / Human Review

任何多候選、approximate flag、label wording mismatch、版本疑義或可影響照護/申報的結論，一律交由 medical coder / neurologist / pharmacist 複核。



Key takeaway

- 固定範圍
- 可追溯來源
- 非自動決策



插入頁 | task_spec.md + scope files

左側以 md 文字呈現規則；右側用表格呈現固定 scope



task_spec.md 摘要



Problem

建立可追溯的固定範圍 ICD crosswalk 與獨立藥物證據 audit。



Input

data/icd9_scope.csv、data/medication_scope.csv、config、scoped source extracts。



Output

crosswalk、drug evidence、source inventory、audit summary、review note、validation report 與 logs。



Rules

不增加 codes/medications；不丟棄候選；不以藥物選 code；不捏造來源；所有 approximate/multiple cases 需 review。



Check

執行 python scripts/validate_outputs.py 與三種 synthetic eval。



Stop

輸入缺漏、來源版本未知、scope 超限、同一錯誤兩輪未改善、或需專業判斷時停止並 handoff。

data/icd9_scope.csv

source_code	source_label_raw	include_in_mvp
345.00	Generalized nonconvulsive epilepsy, not intractable.	TRUE
345.10	Generalized convulsive epilepsy, not intractable.	TRUE
345.40	Localization-related (focal) epilepsy with complex partial seizures.	TRUE
345.90	Epilepsy unspecified, not intractable.	TRUE
345.91	Epilepsy unspecified, with intractable epilepsy.	TRUE

data/medication_scope.csv

medication_id	medication_raw	priority	include_in_mvp
MED001	Phenytoin	1	TRUE
MED002	Carbamazepine	2	TRUE
MED003	Valproic acid	3	TRUE



固定範圍是 validator 的基本前提。



方法與流程

依 hackathon_AGENT_V2.md 分批執行的核心 pipeline



5 個步驟依序執行、產出可追溯且可審查的 crosswalk 與 evidence audit package.



插入頁 | loop_design.md

以 md 文字呈現迴圈式執行與修正規則



Loop Design

- 1 **Plan** : 凍結五碼、三藥、CMS 2018 GEMs 與 FY2026 April release。
- 2 **Act** : 解析、lookup、建立 evidence rows。
- 3 **Observe** : 檢查 row counts、raw flags、來源欄位與 target labels。
- 4 **Evaluate** : 執行 deterministic validator 與 synthetic eval。
- 5 **Revise** : 只修 parser、schema、lookup key、claim scope、status、conversion rule、reference ; 最多 2 輪。
- 6 **Stop** : 全數通過或觸發缺來源、衝突、no progress、專業判斷。

No-progress

連續兩輪相同錯誤。

Human Review triggers

- 多候選 / approximate / context flag
- label mismatch
- target release 疑義
- drug normalization ambiguity
- drug scope 與 subtype 需求不一致



本次實際 revision count = 1



插入頁 | tool_inventory.md + harness_checklist.md

工具清單用表格呈現；檢查項目用 checklist 呈現



Tool inventory

	Tool	Purpose	Input	Output
1	Python csv/text parser	讀取 scope、GEM 與 code extract	CSV/TXT	rows
2	Official crosswalk reader	解析 GEM 固定欄位	scoped GEM TXT	candidates/raw flags
3	ICD-10-CM validator	exact code lookup	candidates + scoped release CSV	status/label
4	Drug terminology lookup	ingredient normalization	medication raw	normalized name + UNII
5	Drug evidence retriever	讀取 DailyMed/MedlinePlus	ingredient	narrow claims
6	Output validator	schema/enums/separation/eval	project files	report/exit code
7	File writer	建立 artifacts	structured rows	CSV/MD

Harness checklist

- ✓ Input schema check
- ✓ Fixed scope check
- ✓ Mapping source version check
- ✓ Raw flag preservation check
- ✓ Candidate completeness check
- ✓ Target code existence check
- ✓ Mapping cardinality check
- ✓ Citation/source trace check
- ✓ Drug name normalization check
- ✓ Drug evidence scope check
- ✓ No drug-to-code inference check
- ✓ Normal case
- ✓ Not-found case
- ✓ Conflicting case
- ✓ Human Review check
- ✓ No fabricated mapping check
- ✓ No fabricated citation check

所有 checklist 項目皆通過。



ICD Crosswalk 結果

5 個 ICD-9-CM source codes, 共保留 11 筆 GEM 候選



Source codes

5



Candidate rows

11



Direct one-to-one

0



Needs human review

5 / 5

ICD-9-CM	原始診斷	保留的 ICD-10-CM 候選	備註
345.00	Generalized nonconvulsive epilepsy	G40.A01, G40.A09	absence epileptic syndrome ; 需區分 status
345.10	Generalized convulsive epilepsy	G40.309, G40.401, G40.409	generalized subtype 多候選
345.40	Focal epilepsy with complex partial seizures	G40.201, G40.209	需區分 status epilepticus
345.90	Epilepsy unspecified, not intractable	G40.901, G40.909	unspecified, 多候選
345.91	Epilepsy unspecified, intractable	G40.911, G40.919	intractable, 多候選



關鍵發現

- 所有候選均完整保留，沒有只取第一筆
- 11/11 rows 的 raw flag = 10000 (approximate = 1)
- 因此本專案輸出為「候選 crosswalk」，不是最終編碼決策



插入頁 | outputs/icd_crosswalk.csv

以表格呈現主要欄位與 11 筆 mapping rows



全部 11 rows
均保留 approximate flag



mapping_id	source_code	target_code	raw_mapping_flags	mapping_cardinality	mapping_class	target_code_status	status	needs_human_review
MAP001	345.00	G40.A01	10000	one_to_many	multiple_candidates	active_in_configured_release	needs_review	TRUE
MAP002	345.00	G40.A09	10000	one_to_many	multiple_candidates	active_in_configured_release	needs_review	TRUE
MAP003	345.10	G40.309	10000	one_to_many	multiple_candidates	active_in_configured_release	needs_review	TRUE
MAP004	345.10	G40.401	10000	one_to_many	multiple_candidates	active_in_configured_release	needs_review	TRUE
MAP005	345.10	G40.409	10000	one_to_many	multiple_candidates	active_in_configured_release	needs_review	TRUE
MAP006	345.40	G40.201	10000	one_to_many	multiple_candidates	active_in_configured_release	needs_review	TRUE
MAP007	345.40	G40.209	10000	one_to_many	multiple_candidates	active_in_configured_release	needs_review	TRUE
MAP008	345.90	G40.901	10000	one_to_many	multiple_candidates	active_in_configured_release	needs_review	TRUE
MAP009	345.90	G40.909	10000	one_to_many	multiple_candidates	active_in_configured_release	needs_review	TRUE
MAP010	345.91	G40.911	10000	one_to_many	multiple_candidates	active_in_configured_release	needs_review	TRUE
MAP011	345.91	G40.919	10000	one_to_many	multiple_candidates	active_in_configured_release	needs_review	TRUE



插入頁 | outputs/audit_summary.csv + outputs/source_inventory.csv

上半部總結 crosswalk / medication ; 下半部列出主要來源與可追溯性



audit_summary.csv

item_type	item_id	candidate_count	mapping_class_or_disease_scope	status	summary
icd_mapping	345.00	2	multiple_candidates	needs_review	候選 G40.A01; G40.A09 , 已全數保留
icd_mapping	345.10	3	multiple_candidates	needs_review	候選 G40.309; G40.401; G40.409 , 已全數保留
icd_mapping	345.40	2	multiple_candidates	needs_review	候選 G40.201; G40.209 , 已全數保留
icd_mapping	345.90	2	multiple_candidates	needs_review	候選 G40.901; G40.909 , 已全數保留
icd_mapping	345.91	2	multiple_candidates	needs_review	候選 G40.911; G40.919 , 已全數保留
medication	MED001	—	generalized_seizure; focal_seizure	supported	只補充 disease scope , 不參與選碼
medication	MED002	—	focal_seizure; generalized_seizure	supported	只補充 disease scope , 不參與選碼
medication	MED003	—	broad_epilepsy	supported	證據較廣泛 , 不能建立 subtype

source_inventory.csv

source_inventory_id	source_name	version_or_revision	access_status	used_for_ids	note
SRC001	CMS 2018 ICD-9-CM to ICD-10-CM GEMs	2018 final GEMs	opened	MAP001-MAP011	官方 release page opened
SRC002	Scoped GEM extract	2018 final GEMs	downloaded	MAP001-MAP011	official ZIP comparison required
SRC003	Scoped ICD-9-CM labels	ICD-9-CM diagnosis set used by 2018 GEMs	downloaded	MAP001-MAP011	both raw and verified wording retained
SRC004	CMS FY2026 ICD-10-CM April 1, 2026 release	April 1, 2026 release	opened	MAP001-MAP011	release metadata opened
SRC005	Scoped FY2026 ICD-10-CM exact lookup file	April 1, 2026 release	downloaded	MAP001-MAP011	only 11 candidate codes
SRC007	DailyMed DILANTIN label	Revised 03/2022	opened	DRUG001;DRUG002	indications only
SRC008	DailyMed TEGRETOL label	Updated 2025-12-16	opened	DRUG003;DRUG004	indications only
SRC009	MedlinePlus Valproic Acid	Last revised 2026-04-15	opened	DRUG005	broad seizure statement only



重點：每一筆結果都能回查到來源與用途。



Medication Evidence 結果

3 種藥物，整理出 5 筆支持性證據



Medications

3



Evidence rows

5



Rule

藥物證據不參與 ICD 選碼

Medication	Normalized	Disease scope	來源	說明
Phenytoin	Phenytoin	generalized_seizure; focal_seizure	DailyMed	支持 tonic-clonic 與 psychomotor/temporal lobe seizures
Carbamazepine	Carbamazepine	focal_seizure; generalized_seizure	DailyMed	支持 partial complex 與 generalized tonic-clonic seizures
Valproic acid	Valproic acid	broad_epilepsy	MedlinePlus	僅能保守支持 certain types of seizures



Interpretation

- Phenytoin 與 carbamazepine 可支持 focal / generalized 類別
- Valproic acid 本次僅保守標記為 broad_epilepsy
- 藥物 evidence 只能補充 disease scope，不能反推 ICD code



插入頁 | outputs/drug_evidence.csv

以表格呈現藥物證據列、來源與限制，並和 ICD 選碼明確分離



Scoped medications

3



Evidence rows

5



Supported

5 / 5



Needs human review

3 / 5

evidence_id	drug	normalized / UNII	disease_scope	source	status	limitation
DRUG001	Phenytoin	Phenytoin / UNII:6158TKW0C5	generalized_seizure	DailyMed DILANTIN	supported	tonic-clonic evidence only, not an ICD subtype
DRUG002	Phenytoin	Phenytoin / UNII:6158TKW0C5	focal_seizure	DailyMed DILANTIN	supported	older label wording preserved; no ICD candidate selection
DRUG003	Carbamazepine	Carbamazepine / UNII:33CM23913M	focal_seizure	DailyMed TEGRETOL	supported	seizure category only, not code assignment
DRUG004	Carbamazepine	Carbamazepine / UNII:33CM23913M	generalized_seizure	DailyMed TEGRETOL	supported	mixed patterns noted; no patient inference
DRUG005	Valproic acid	Valproic acid / UNII:614OI1Z5WI	broad_epilepsy	MedlinePlus	supported	broad seizure statement only; subtype unspecified



關鍵發現



Phenytoin 與 Carbamazepine 各有 focal / generalized 分列



Valproic acid 僅保守分類為 broad_epilepsy



藥物證據不參與 ICD-10-CM 候選選碼



重點：藥物只提供 disease scope 證據，不直接決定 ICD crosswalk。



Harness 驗證結果

3 組 synthetic fixtures 全數通過，驗證安全失敗衝突升級



Eval cases

3



Passed

3 / 3



Needs human review

2 / 3



Validator checks

38 / 38



data/eval_cases.csv

eval_id	case_type	source_code	provided_target_code	medication_raw	expected_status
EVAL001	normal	345.00	—	Phenytoin	needs_review
EVAL002	not_found	INVALID_ICD9_TEST	—	UNKNOWN_MEDICATION	not_found
EVAL003	conflicting	345.90	G40.A01	—	conflicting



outputs/eval_results.csv

eval_id	observed_status	needs_human_review	passed	result
EVAL001	needs_review	TRUE	TRUE	retained both GEM candidates; drug evidence stayed separate
EVAL002	not_found	FALSE	TRUE	invalid code and unknown medication produced no fabricated output
EVAL003	conflicting	TRUE	TRUE	supplied G40.A01 conflicted and was escalated



Safety behavior

- normal case：保留候選，不做最終選碼
- not_found：安全失敗，不捏造 target code 或 drug mapping
- conflicting：保留原始衝突並升級 Human Review



插入頁 | outputs/validation_report.md

以 md 文字呈現 validator 執行結果與 revision 記錄



outputs/validation_report.md

Validation Report

Command: python scripts/validate_outputs.py

Exit status: 0

Passed checks: 38

Failed checks: 0

1. Scope & mapping

- Required output files exist
- Fixed scope exactly five
- All five codes represented
- Candidate completeness against frozen extract 11 / 11
- Raw flags preserved
- Target exact lookup present
- All ambiguous mappings reviewed

2. Drug evidence

- Medication limit
- Drug evidence row limit
- Drug evidence IDs unique
- Supported drug evidence complete
- No ICD fields in drug evidence
- No drug-to-code inference

3. Audit & eval

- Source inventory categories
- Local source files hashed
- Audit summary coverage
- Eval cases synthetic
- Normal / Not-found / Conflicting fixtures passed
- Conflict escalated
- No patient-level diagnosis or recommendation

4. Revision

- Revision count 1
- Revision 1 corrected the scoped ICD-10 lookup label-field name after a KeyError
- CMS binary ZIP limitation remains a Human Review item



Summary

38

passed

0

failed



Key takeaway

- Validator 全數通過
- 問題已修正 1 次
- 剩餘限制已明示，不隱藏



插入頁 | outputs/review_note.md

以 md 文字呈現需要人工複核的原因、限制與 reviewer 角色



outputs/review_note.md 摘要

1. ICD mapping

All five source codes require review because each has multiple target candidates and every retained GEM row has raw flag 10000 (approximate=1). No unique target was selected.

345.00 -> G40.A01; G40.A09	reason: status epilepticus distinction and approximate flag	reviewer: Medical coder / neurologist
345.10 -> G40.309; G40.401; G40.409	reason: generalized subtype and status distinction	reviewer: Medical coder / neurologist
345.40 -> G40.201; G40.209	reason: status epilepticus distinction	reviewer: Medical coder / neurologist
345.90 -> G40.901; G40.909	reason: status epilepticus distinction	reviewer: Medical coder
345.91 -> G40.911; G40.919	reason: status epilepticus distinction	reviewer: Medical coder

2. Drug evidence

Phenytoin and carbamazepine support focal / partial and generalized categories in separate rows. Valproic acid was conservatively classified broad_epilepsy. None of these rows can determine an ICD candidate.

3. Source-package handoff

Official CMS release metadata and the GEM user guide were opened, but ZIP binaries could not be retained in the sandbox. Compare the scoped rows and target-code extract with official packages before operational use.

Recommended reviewer roles



Health information manager / medical coder



Neurologist



Pharmacist



Data steward



Human Review triggers



multiple candidates



approximate flag



wording mismatch



broad drug evidence



Thanks



插入頁 | demo_notes.md + MANIFEST.csv

左側為 5 分鐘 demo 節點；右側為封存與交付清單



demo_notes.md 摘要

- 0:00-0:40 Problem —**
ICD-9-CM to ICD-10-CM is not assumed one-to-one.
Drug evidence is not coding evidence.
This is an evidence audit, not automatic decision-making.
- 0:40-1:20 Scope and Sources —**
Show the five fixed codes, CMS 2018 GEMs,
FY2026 April target release,
and the three scoped ingredients.
- 1:20-2:30 Crosswalk —**
Use 345.10 as the example: three candidates are retained.
Show 10000, one-to-many cardinality,
target exact lookup, and needs_review.
- 2:30-3:20 Drug Evidence —**
Show separate focal/generalized rows for carbamazepine
and the conservative broad row for valproic acid.
No medication row selects an ICD code.
- 3:20-4:15 Harness —**
Show normal, not_found, and conflicting synthetic cases
plus validator checks.
- 4:15-5:00 Limits and Handoff —**
No diagnosis, final coding, prescribing, dose,
prognosis, or patient advice.
All ambiguous mappings go to a coder / neurologist;
source ZIP extracts require independent verification.

MANIFEST.csv highlights

path	size_bytes	sha256 (short)
data/eval_cases.csv	435	116d9318...
outputs/drug_evidence.csv	2934	76082d1c...
outputs/icd_crosswalk.csv	9625	9032839e...
outputs/validation_report.md	1550	1629686f...
logs/run_log.md	2172	1cadc221...



所有主要輸出皆已封存並可驗證雜湊。

Final handoff

- 交付 zip package + MANIFEST
- 保留 review_note 與 validation_report
- 可做 demo / audit, 不可直接做最終臨床或申報決策。



結論：本專案完成可追溯 crosswalk 與 evidence audit package, 但所有模糊結果皆需人工複核。